



Virtual GlobeTrotter

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Team Role Distribution

Viola



Scrum Master
API
UI
Trello
Connectivity

Atvars



Development
AI
Unity
Connectivity

Isabelle



Product Owner
Database
AI
Documentation
Connectivity

Project Description



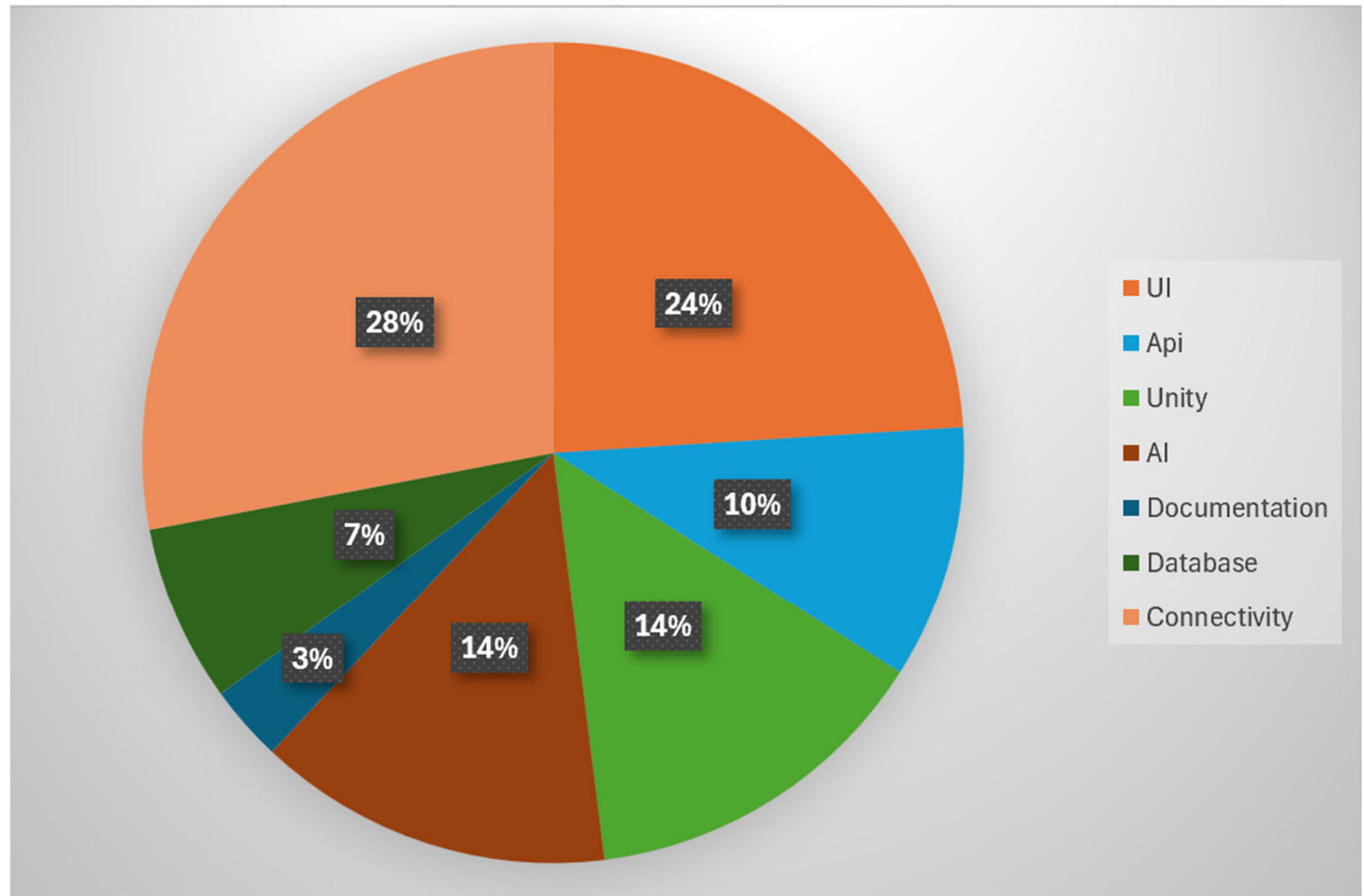
What? 360° AI generated immersion based on voice input to simulate a study trip.

For? Children around 10 yrs

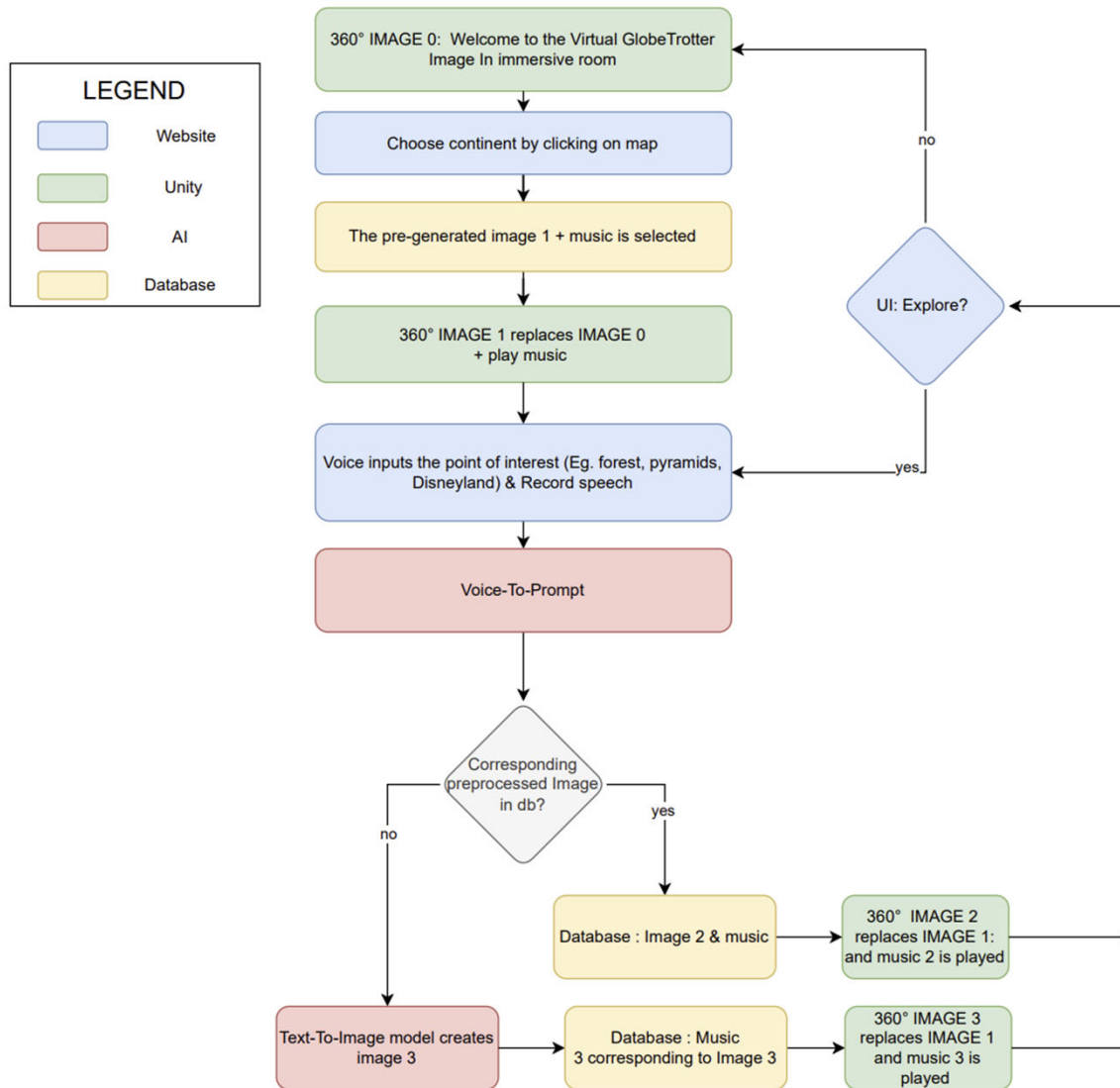
Where? Immersive Room in Forum Core Howest

- Tablet with **user interface + world map**
- **Continent choice** and **voice input**.
- A **360° image** is shown and **music** is played.
- The image and music is selected in a database of **pregenerated** images and music or a **new** image is created by an AI model. The image corresponds to the voice input, the music to the chosen continent

Story points



Technical solution Diagram

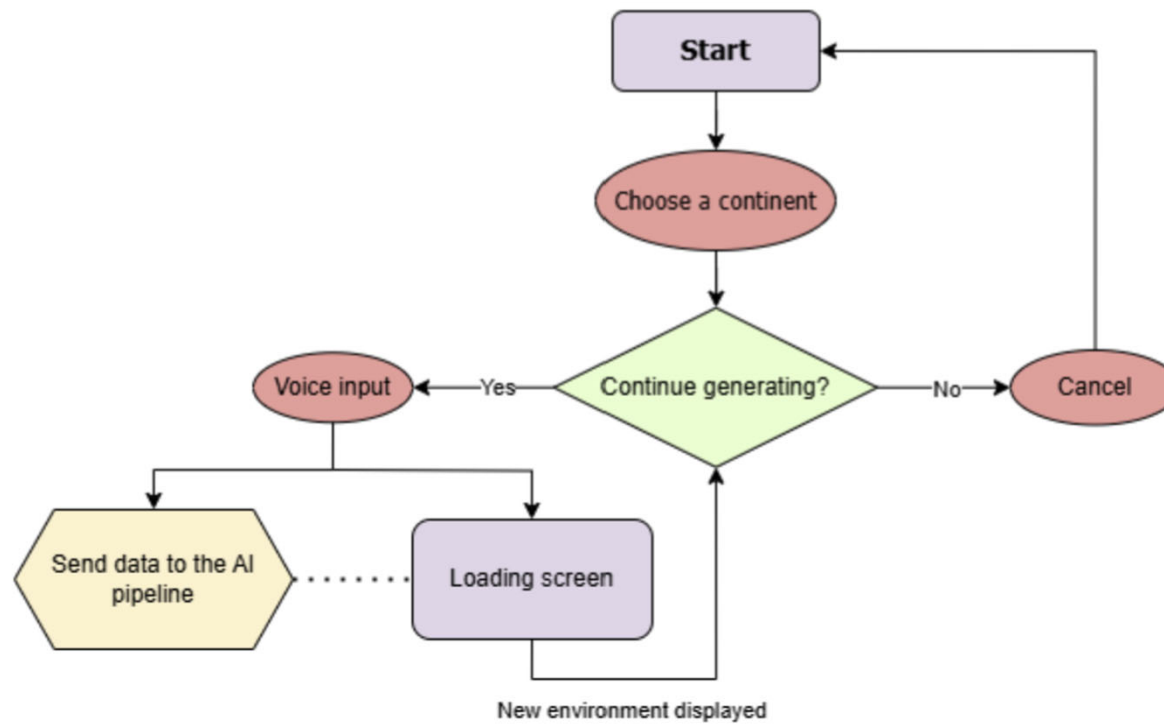


Website: The Start

- Flask for API: established connection with the database and AI class.
 - Vue.js for Frontend
 - Docker container
 - Accessible through a local tablet
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- Easy to navigate
 - Visuals corresponding to the project concept
 - Connection point for all parts of the project



User Journey



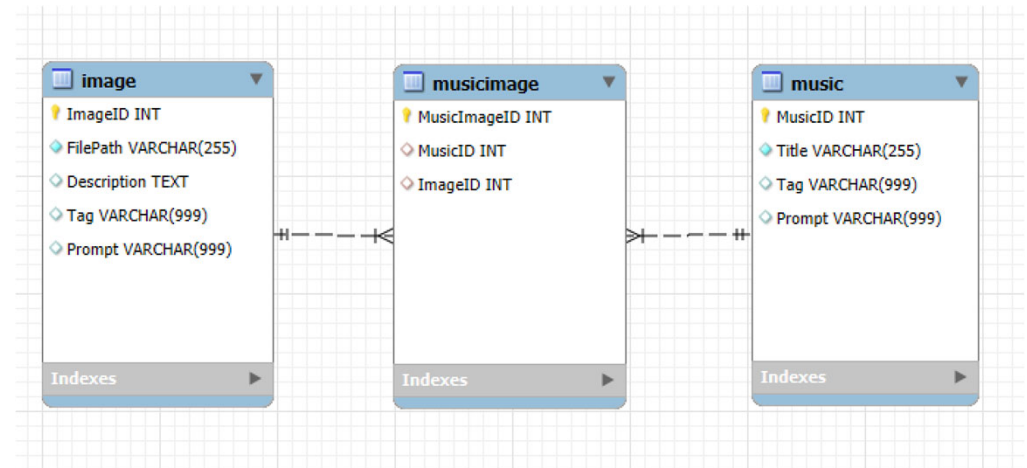
Database: find the right one



The database :

- path to the generated images
- prompt used to generate the images by the AI model.
- 5 Music / sounds by continent

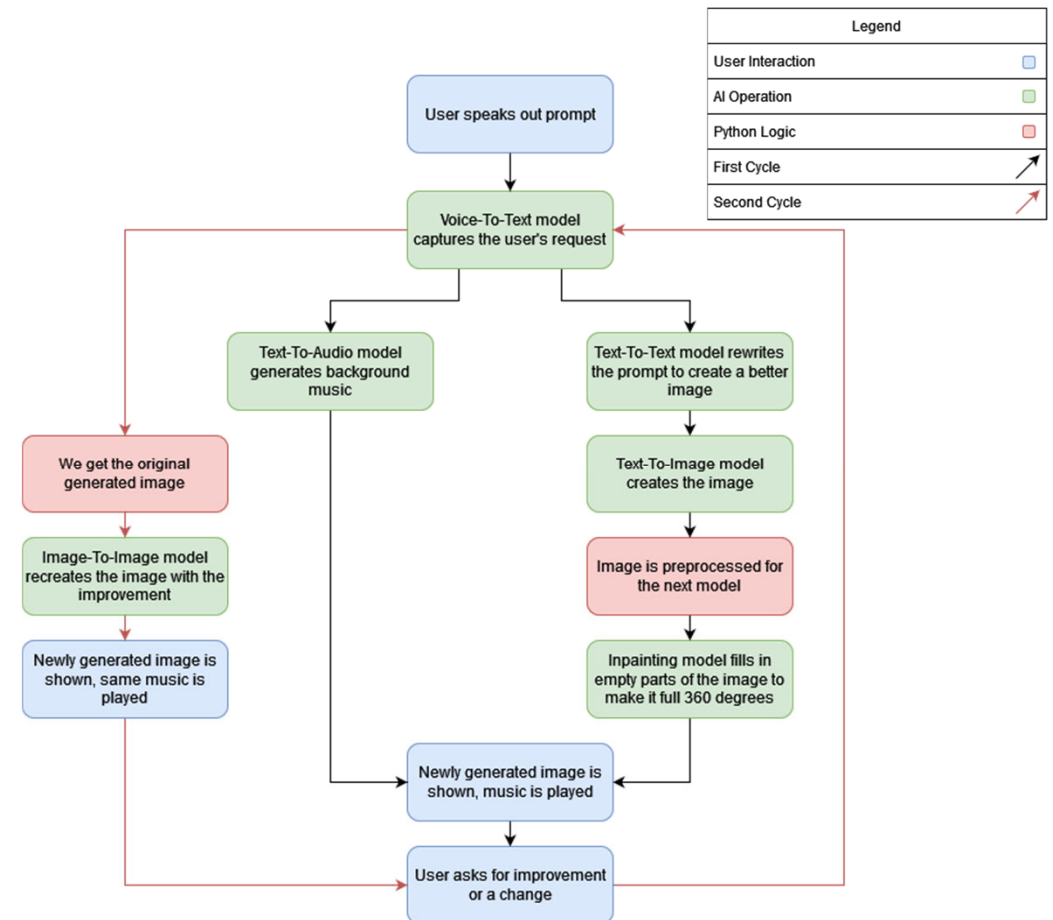
1st idea : Relational Database / can be NoSql



AI: and else create it!

AI Pipeline:

- Voice-To-Text
 - Converts user input from audio to text
- Text-To-Audio
 - Creates music from user prompt
 - Undecided yet whether real-time or pre-generated
- Text-To-Text
 - Prompt enhancing
- Text-To-Image
 - Creating the environment
- Inpainting
 - Removing the seam
- Image-To-Image
 - Adding user requested improvements to image



Unity: Ears and Eyes ready?

- Needed for interacting with the Immersive room
- Empty environment with a giant sphere, user is in the centre
- Image overlaid on inside surface of the sphere
- AI generated music playing in the background



Biggest Challenge!

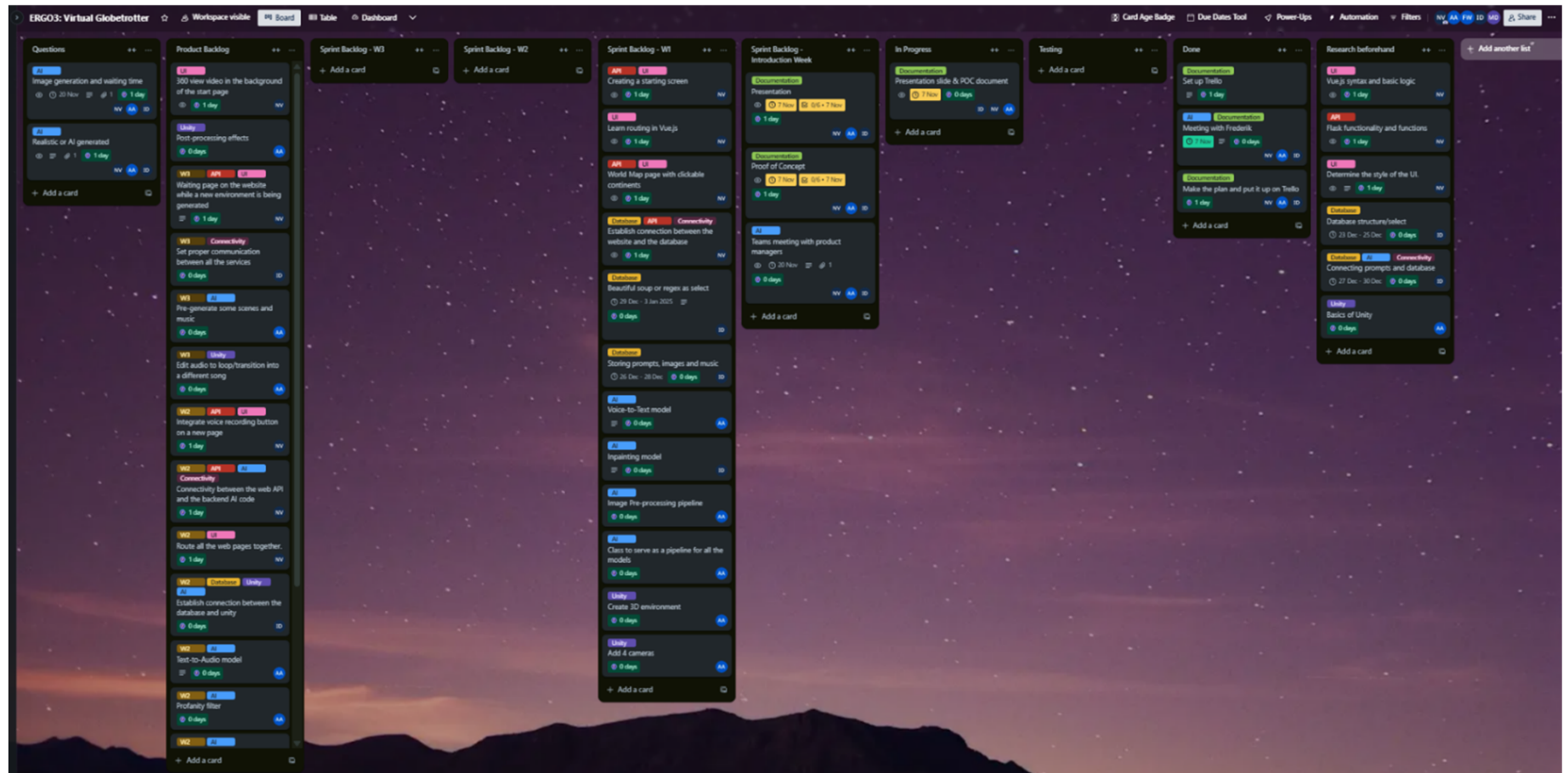
Connecting everything to 1 experience: is the biggest challenge!

Examples of connectivity

- connecting AI model to the database: path to images and prompt / keywords
- database to unity to show an image or the play music in the immersive room



Trello



Trello

